

$$B) \quad n_{\text{ograd}} = \frac{\partial L}{\partial n} \quad L = \tanh(n)$$

$$\frac{\partial L}{\partial n} = \text{slope} = \frac{f(n+h) - f(n)}{h}$$

$$= \frac{\tanh(n+h) - \tanh(n)}{h}$$

we know that,

→ skipping calc.
moving
to cal.

$$\tanh(n) = \frac{e^n - e^{-n}}{e^n + e^{-n}}$$

$$\sinh(n) = (e^n - e^{-n}) / 2$$

$$\cosh(n) = (e^n + e^{-n}) / 2$$

$$\frac{\partial \tanh(n)}{\partial n} = \frac{\partial}{\partial n} \left(\frac{e^n - e^{-n}}{e^n + e^{-n}} \right)$$

B) (calculus, $\frac{\partial L}{\partial n} = \partial_1 m \cdot g^{\text{rad}},$
 $L = \tanh(n)$)

$$\frac{\partial \tanh(n)}{\partial n} = \frac{\partial (e^n - e^{-n})}{\partial n (e^n + e^{-n})}$$

(we know that,
 $(d(x/y) = x dy + y dx)$)

$(d(x/y) = \frac{y dx - x dy}{y^2})$

$$\frac{\partial \tanh(n)}{\partial n} = \frac{(e^n + e^{-n}) \partial (e^n - e^{-n}) - (e^n - e^{-n}) \frac{\partial (e^n + e^{-n})}{\partial n}}{(e^n + e^{-n})^2}$$

(we know that, $d(e^x) = e^x$)
 $(e^{-x} = -e^{-x})$

so, $\partial (e^n - e^{-n}) = (e^n + e^{-n})$

$\partial (e^n + e^{-n}) = (e^n - e^{-n})$

$$\frac{\partial}{\partial n} (\tanh(n)) = \frac{(e^n + e^{-n})(e^n + e^{-n}) - (e^n - e^{-n})(e^n - e^{-n})}{(e^n + e^{-n})^2}$$

$$= \frac{(e^n + e^{-n})^2 - (e^n - e^{-n})^2}{(e^n + e^{-n})^2}$$

$$= \left(\frac{(e^n + e^{-n})^2}{(e^n + e^{-n})^2} - \frac{(e^n - e^{-n})^2}{e^n + e^{-n}} \right)^2$$

$$= 1 - \left(\frac{e^n - e^{-n}}{e^n + e^{-n}} \right)^2$$

$$\frac{\partial \tanh(n)}{\partial n} = 1 - (\tanh(n))^2 \quad \rightarrow \text{calculus}$$

$$\boxed{\frac{\partial L}{\partial n} = n.\text{grad} = (1 - \tanh(n))^2}$$

$$= (1 - L^2) \quad (L = \tanh(n))$$

$$= (1 - (0.7071)^2)$$

$$= 0.5$$

back to slope

$$\frac{\partial L}{\partial n} = \frac{1}{1-L^2} \quad L = \tanh(n)$$

$$h \rightarrow 0 \Rightarrow h \approx 0$$

$$\text{slope} = \frac{f(n+h) - f(n)}{h} = \frac{\tanh(n+h) - \tanh(n)}{h}$$

(we know that,

$$\tanh(A+B) = \frac{\tanh(A) + \tanh(B)}{1 + \tanh(A)\tanh(B)})$$

$$= \frac{\tanh(n) + \tanh(h)}{1 + \tanh(n)\tanh(h)} - \tanh(n)$$

$$\lim_{h \rightarrow 0} \frac{1 + \tanh(n)\tanh(h)}{h}$$

$$= \left(\frac{L + \tanh(h)}{1 + L \tanh(h)} - L \right)$$

$$= \frac{L + \tanh(h)}{1 + L \tanh(h)} - L$$

$$= \frac{L + \tanh(h) - L(1 + L \tanh(h))}{h(1 + L \tanh(h))}$$

$$= \frac{\cancel{L} + \tanh(h) - \cancel{L} - L^2 \tanh(h)}{h(1 + L \tanh(h))}$$

$$= \frac{\tanh(h)(1 - L^2)}{h(1 + L \tanh(h))}$$

$$= \lim_{h \rightarrow 0} \left(\frac{\tanh(h)}{h} \right) (1 - L^2) \left(\frac{1}{1 + L \tanh(h)} \right)$$

(we know that $\lim_{h \rightarrow 0} \frac{\tanh(h)}{h} = 1$)

$$\left(\frac{1}{1 + L \tanh(h)} \right) \xrightarrow{h \rightarrow 0} 1$$

$$\frac{\partial L}{\partial n} = (1 - L^2) = 1 - (0.7)^2 = (0.5)$$

→ Analytically